

① Hashing Function

To save spaces and time each record stored in a computer is assigned an address in computer's memory. The task of assigning addresses is performed by Hashing function

$$H: K \xrightarrow{1-1} L$$

onto.

K - set of keys

L $\rightarrow$ memory addresses

Hashing Function is a means of identifying record which is one to one. There are 3 methods for finding Hash functions

- ① Division Method.
- ② Mid-Square Method.
- ③ Folding Method.

Q Consider an organisation which has 53 employees and each one is assigned a unique 3 digit employee number. Suppose the list consist of 100 2 digit addresses (00, 01, ..., 99). Determine the Hash addresses generated by the Hash function

Congruence Modulo

$$h(x) \equiv x \pmod{97} \text{ for the employee numbers}$$

x	$h(x)$
934	$934/97 \rightarrow 61$
971	$\rightarrow 01$
108	$\rightarrow 11$
902	$\rightarrow 29$
632	$\rightarrow 50$

* If remainder repeats
 $\rightarrow$ collision occurs

Successive no. is assigned in such case.

provided not in use already

$$a \equiv b \pmod{m}$$

$\rightarrow$ a is congruent

Q For Following hash function show that the data would be inserted in the order given in initially empty cells. Use the usual collision resolution policy to resolve collisions, if occur.

$$h(x) = x^2 \pmod{11}$$

Cells are indexed from 0 to 10.

The data is 53, 13, 281, 743, 377, 20, 10, 796.

x	x^2	$h(x) = x^2 \pmod{11}$	$h(x)$
53	2809	4	4
13	169	④ → collision → ⑤	5
281	78961	3	3
743	552049	③ → collision → ⑥	6
377	142129	9	9
20	400	④ → collision → ⑦	7
10	100	1	1
796	633616	⑤ → collision → ⑧	8

Q Using Hashing function which takes first 3 digits of a 7 digit no. as one number and last four digits as another no, adds them and then apply $(\pmod{59})$ function, determine the hash addresses for the following numbers.

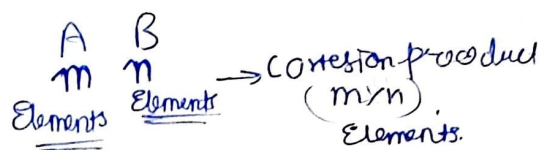
	x	$h(x) \equiv x \pmod{57}$
→ 375 9273	9648	31
714 9021	9735	0
516 7249	7765	36
256 1384	1640	47
608 2376	2984	34

Q For the following hashing function show how the given data would be inserted in order even in initially empty cells if $h(x) \equiv (x^2 + x) \pmod{17}$. Cells are indexed from 0 to 16 and the data is

		$h(x) \equiv (x^2 + x) \pmod{17}$
14	210	6
631	398792	⑥ → ⑦
26	702	5
373	139502	0
775	601400	8
906	821742	13
509	259590	① → ①
2032	4131056	⑤ → ⑨
42	1806	2
4	20	3
136	18632	① → ②
1028	1057812	4 → ⑩

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Binary Relation



Let A, B be any two sets, then a Relation R from $A \rightarrow B$ is a subset of $A \times B$ (Cartesian product).
 $R \subseteq A \times B$.

$$R = \{ (a, b) : a \in A, b \in B, a R b \}$$

If R is a subset of $A_1 \times A_2 \times A_3 \dots A_n$ then R is called n -array Relation

With reference to it
 $R \subseteq A_1 \times A_2$
 is binary Relation.

Relation :

Function :

$$\begin{array}{cc} F \rightarrow R & \checkmark \\ R \rightarrow F & \times \end{array}$$

(Trick)

Function can be many one
not one many

Relation on a set (Not on binary Relation)

$$R \subseteq A \times A$$

R is said to be a relation on set A if $R \subseteq A \times A$.

properties of Relation on a set

① Reflexive :

(For all)

if $a R a, \forall a \in A$
 or
 $(a, a) \in R$

[parallel, $\leq$, $\geq$...]

② Symmetric

If $a R b \Rightarrow b R a, a, b \in R.$

"Jitne hai unke liye"

③ Antisymmetric (one-sided Relation)

If $a R b$ and $b R a \Rightarrow a = b$ — ①
$$P \longrightarrow Q$$

$$\equiv \sim Q \rightarrow \sim P$$

If $a \neq b \Rightarrow a \nrightarrow b$ or $b \nrightarrow a$ — ②
$$\downarrow \qquad \downarrow$$
$$b R a \qquad a R b$$

④ Transitive

If $a R b$ and $b R c \Rightarrow a R c.$

Eg - x is taller than y .

$$R \subseteq A \times B.$$

Q List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$
 $B = \{0, 1, 2, 3\}$

where $(a, b) \in R$ iff

① $a = b$

② a/b (divides) (a divides b)

③ $\gcd(a, b) = 1$

① $R = \{(0,0) (1,1) (2,2) (3,3)\}$

② $R = \{(1,0) (1,1) (1,2), (1,3), (2,0) (3,3) (3,0) (3,3) (4,0)\}$

③ $R = \{(1,1) (1,2) (1,3) (2,1) (2,3) (3,1) (3,2) (4,1) (4,3)\}$

Q Which of these relations on set A are Reflexive, symmetric, anti-symmetric or transitive.

Soln $A = \{1, 2, 3, 4\}$

① $\{(2,2)(2,3)(2,4)(3,2)(3,3)(3,4)\}$
 $R \times \quad S \times \quad AS \times \quad T \checkmark$ $(2,2)(2,3)(3,2)$
 $(3,2)(2,3)(3,3)$

② $\{(1,1)(2,2)(3,3)(4,4)\}$

$R \checkmark$
 $S \checkmark$ (using multiplicity of sets).
 $AS \checkmark$ $1R1$ and $1R1 \Rightarrow 1=1$.
 $T \checkmark$ $(1,1)(1,1) \rightarrow (1,1)$

③ $\{(1,3)(1,4)(2,3)(2,4)(3,1)(3,4)\}$

$R \times$
 $S \times$
 $AS \times$
 $T \times$

Q Determine whether the Relation R defined on the set of Integers is Reflexive, symmetric, ^{AS}transitive or not?

① $x \neq y$

② $x \equiv y \pmod{7}$

③ $x = y^2$

② or $x \neq y/7$

① $x \neq y$
 $R \times$
 $S \checkmark$
 $AS \times$
 $T \times$

① $R: x - x/7$
 $\rightarrow x \equiv x \pmod{7}$

② $S: x - y/7 \Rightarrow -(x - y)/7$
 $= (y - x)/7$
 $\Rightarrow y \equiv x \pmod{7}$

③ $A: x - y/7 \wedge y - z/7$

$$x-y = 7k_1 \wedge y-z = 7k_2$$

$$\therefore x-z = x-y+y-z = 7k_1+7k_2$$

$$\Rightarrow 7(k_1+k_2) \\ = 7k$$

$$\therefore x-z \div 7$$

$$\textcircled{3} \quad x=y^2$$

$$R \quad X$$

$$S \quad X$$

$$AS \quad \checkmark$$

$$T \quad \checkmark$$

$$\textcircled{1} \quad x=x^2 \quad \forall x \quad X$$

$$\textcircled{2} \quad x=y^2 \Rightarrow y=x^2 \quad X$$

$$\textcircled{3} \quad \text{second definition holds}$$

$$\textcircled{4} \quad x=y^2 \wedge y=z^2 = x=z^4 \dots$$

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$$A = \{1, 2, 3\}$$

① Inverse Relation:

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

$$R = \{(1, 1) (1, 2) (2, 3)\}$$

$$R^{-1} = \{(1, 1) (2, 1) (3, 2)\}$$

② Complementary Relation:

$$\bar{R} = \{(a, b) : (a, b) \notin R\}$$

$$\bar{R} = \{(1, 3) (2, 1) (2, 2) (3, 1) (3, 2) (3, 3)\}$$

$$R \subseteq A \times A$$

we take the remaining from universal set. For $\bar{R}$.

③ Combining Relations:

$$A = \{1, 2, 3\}$$

$R_1, R_2 \rightarrow \text{relations}$

$$\textcircled{1} \quad R_1 \cup R_2$$

$$R_1 = \{(1, 1) (1, 3) (3, 2) (3, 3)\}$$

$$\textcircled{2} \quad R_1 \cap R_2$$

$$R_2 = \{(1, 2) (1, 3) (2, 3) (3, 2)\}$$

$$\textcircled{3} \quad R_1 - R_2$$

$$R_1 \cup R_2 = \{(1, 1) (1, 2) (1, 3) (2, 3) (3, 2) (3, 3)\}$$

$$\textcircled{4} \quad R_2 - R_1$$

$$R_1 \cap R_2 =$$

$$R_1 - R_2 =$$

$$R_2 - R_1 =$$

Composite of Relations

$$R \subseteq A \times B \rightarrow \{(a,b) : a \in A, b \in B\}$$

$$S \subseteq B \times C \rightarrow \{(b,c) : b \in B, c \in C\}$$

$$S \circ R \subseteq A \times C$$

$$= \{(a,c) : (a,b) \in R \wedge (b,c) \in S\}$$

R Pehle

S baad mein.

$$R \circ S \subseteq A \times C$$

$$\{(c,a) : (a,b) \in S \wedge (b,c) \in R\}$$

S Pehle /

R baad mein.

Powers of Relation

$$R = \{(1,1), (1,2), (2,1)\}$$

$$R = \{(1,1), (1,2), (2,1)\}$$

$$R^2 = R \circ R = \{(1,1), (1,2), (2,1), (2,2)\}$$

$$R^3 = R \circ R^2$$

$$R^4 = R^3 \circ R^2$$

$$R_1 \oplus R_2 = (R_1 \cup R_2) - (R_1 \cap R_2)$$

Q Let R be the Relation $R = \{(a,b) : a < b\}$ on $\mathbb{I}$
Then Evaluate.

$$\textcircled{1} R^{-1} = \{(a,b) : a > b\}$$

$$\textcircled{2} \bar{R} = \{(a,b) : a \geq b\}$$

Q $R = \{(a, b) : a \text{ divides } b\} \quad \left(\frac{b}{a}\right)$

① $R^{-1} = \{(a, b) : b \text{ divides } a\} \quad \left(\frac{a}{b}\right)$

② $\bar{R} = \{(a, b) : a \text{ does not divide } b\}$

Q3 $R_1 = \{(1, 2) (2, 3) (3, 4)\}$

$R_2 = \{(1, 1) (1, 2) (2, 1) (2, 2) (2, 3) (3, 1) (3, 2) (3, 3) (3, 4)\}$

$(R_1 \subset R_2) \rightarrow \text{by observation.}$

① $R_1 \cup R_2 = R_2$

② $R_1 \cap R_2 = R_1$

③ $R_1 - R_2 = \emptyset$

④ $R_2 - R_1 = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3)\}$

Q4 $R = \{(1, 2) (1, 3) (2, 3) (2, 4) (3, 1)\}$

$S = \{(2, 1) (3, 1) (3, 2), (4, 2)\}$

$\hookrightarrow S \circ R = \{(1, 1) (1, 2) (2, 2) (2, 1)\}$

Q Given the following Relations on the set of Real numbers R

① $R_1 = \{(a, b) : a > b\}$

② $R_6 = \{(a, b) : a \neq b\}$

③ $R_2 = \{(a, b) : a \geq b\}$

Find ① $R_2 \cup R_4$

⑤ $R_3 - R_6$

④ $R_3 = \{(a, b) : a < b\}$

② $R_3 \cup R_6$

⑥ $R_6 - R_3$

⑤ $R_4 = \{(a, b) : a \leq b\}$

③ $R_3 \cap R_6$

⑦ $R_2 \oplus R_6$

⑥ $R_5 = \{(a, b) : a = b\}$

④ $R_4 \cap R_6$

⑧ $R_3 \oplus R_5$

Soln ① $R_2 \cup R_4 = R^2$ (as we are dealing with pairs).

unusual set.

② $R_3 \cup R_6 = R_6$

③ $R_3 \cap R_6 = R_3$
 $\langle \langle \rangle \rangle$



④ $R_4 \cap R_6 \Rightarrow R_3$
 $\langle = \rangle <$

⑤ $R_3 - R_6 = \emptyset$

⑥ $R_6 - R_3 = R_1$

⑦ $R_2 \oplus R_6 \Rightarrow (R_2 \cup R_6) - (R_2 \cap R_6) = R^2 - R_1 = R_4$

⑧ $R_3 \oplus R_5 = (R_3 \cup R_5) - (R_3 \cap R_5)$
 $\Rightarrow R_4 - \emptyset = R_4$

• Representing Relations

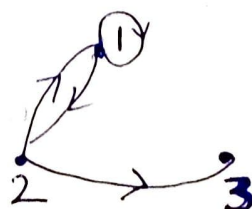
① Ordered pair.

② Matrix.

③ Digraphs $\rightarrow$ vertices are the elements of set A.

$$A = \{1, 2, 3\}$$

$$R = \{(1,1) (1,2) (2,1) (2,3)\}$$



① Reflexive - loops

② Symmetric - opp edges

③ Anti-symmetric - only one edge/connection

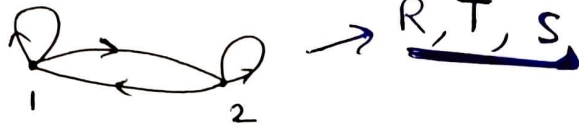
④ Transitive. (check kar)

$\hookrightarrow (a,b) \wedge (b,c) \Rightarrow (a,c)$
 edge.

- If two edges, then two loops are must for Transitive then check for others also

Q Check whether the relation R defined on set $A = \{1, 2, 3, 4\}$ represented by digraphs are R, S, T or AS.

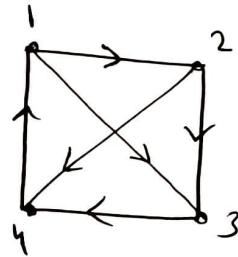
①



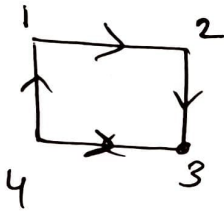
②



③



④



↳ Asymmetric

↳ Asymmetric.

— X —